Sentiment Analysis Report

1. Dataset Description

The dataset used in this project is the Datafiniti Consumer Reviews of Amazon Products collection from Kaggle.
It contains verified customer reviews for a wide range of Amazon products, including electronics, household items, clothing, and more.

Key characteristics of the dataset:

- Contains product reviews written by Amazon customers.
- Includes fields such as:
    - *reviews.text* – the main review content
    - *reviews.rating* – the customer rating
    - *products.title* – product name
    - *reviews.date* – date the review was posted
- The dataset is supplied as a large CSV file with duplicated and incomplete entries, which makes data cleaning essential.

2. Pre-processing Steps

The following steps were applied to prepare the data for sentiment analysis:

a. Handling Missing and Duplicate Reviews

- Removed rows where *reviews.text* was null.
- Removed duplicated reviews.
- Output included a summary of how many rows were removed.

b. Text Normalisation

Each review was processed to improve consistency and model performance:

- Converted all text to lowercase.
- Stripped leading and trailing whitespace.
- Tokenised text using spaCy.
- Removed:
    - Stop words (e.g., *the, is, of*)
    - Punctuation marks
- Joined the cleaned tokens back into a normalised text string.

c. Additional NLP Processing

The *spacytextblob* extension was added to the spaCy pipeline to enable:

- Polarity scoring (sentiment between –1 and +1)
- Subjectivity scoring (0 = objective, 1 = opinion-heavy)

3. Evaluation of Results

Sentiment Classification

Each review received:

- A **sentiment label** (“Positive”, “Negative”, or “Neutral”) based on polarity thresholds.
- A corresponding **polarity** and **subjectivity** score.

Example results from sample reviews demonstrated:

- Clearly **positive reviews** scored **high polarity (0.3833).**
- Clearly **negative reviews** scored **low polarity (-0.875).**
- **Neutral** or mixed reviews scored **near zero (0.0).**

Similarity Comparison

The first two reviews from the dataset were compared using ‘spaCy’ embeddings:

- A numerical similarity score (0–1) was generated.
- **Higher values** indicate more **semantic overlap (0.8651).**
- This helped evaluate whether different customers expressed similar opinions about products.

Data Output

The final dataframe contained new columns:

- clean_text
- sentiment_label
- polarity
- subjectivity

A preview of the first 10 entries confirmed correct processing.

4. Strengths and Limitations of the Model

Strengths

- **Simple and interpretable sentiment model:** 'TextBlob' polarity scores are easy to understand and explain.
- **Efficient pre-processing:** 'spaCy' handles tokenisation and stop-word removal robustly.
- **Strong performance on clear sentiment:** Positive or negative reviews with emotional language are accurately classified.
- **Built-in subjectivity scoring:** Helps distinguish factual reviews from opinion-heavy content.

Limitations

- **Domain sensitivity:** 'TextBlob' is not fine-tuned for Amazon reviews specifically, so subtle sentiment may be misclassified.
- **Limited nuance:**
  Sarcasm, mixed opinions, and context-heavy sentences may produce inaccurate scores.
- **No use of deep learning sentiment models:** More modern transformer-based models (e.g., BERT) would outperform TextBlob.
- **SpaCy medium model embeddings:**
  en_core_web_md provides good similarity estimates but lacks the semantic depth of transformer models like *en_core_web_trf.*

Summary

This project successfully implemented a complete sentiment analysis pipeline using spaCy, TextBlob, and Kaggle data. The system loaded, cleaned, preprocessed, analysed, and compared product reviews, producing interpretable sentiment scores and similarity metrics. While the approach is lightweight and effective for general sentiment tasks, it has limitations with nuanced language and could be improved by using more advanced transformer-based NLP models.

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